

2024 Linear Algebra and Calculus Resit

- ① Evaluate the following limit: $\lim_{x \rightarrow \infty} \frac{x^3 + 3x^2 + 2}{2x^4 + 5x^3}$.

Taking dominant terms $\lim_{x \rightarrow \infty} \frac{x^3 + 3x^2 + 2}{2x^4 + 5x^3} = \lim_{x \rightarrow \infty} \frac{x^3}{2x^4} = \lim_{x \rightarrow \infty} \frac{1}{2x} = 0$.

- ② Show that $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$.

Since $-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$, $x^2(-1) \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2(1) \rightarrow -x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2$. Since $\lim_{x \rightarrow 0} -x^2 = \lim_{x \rightarrow 0} x^2 = 0$, by the Squeeze Theorem, $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$.

Alternatively, one can simply argue that since $\sin\left(\frac{1}{x}\right)$ is bounded by -1 and 1 , $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = \underbrace{\lim_{x \rightarrow 0} x^2}_0 \lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right) = 0$.

- ③ Evaluate the integral $I = \iint_R x e^{-y} dy dx$

where R is the region defined by $0 \leq x \leq 2$ and $y \geq 0$.

The integral over R is then $I = \int_0^2 \int_0^\infty x e^{-y} dy dx = \int_0^2 x dx \int_0^\infty e^{-y} dy$. First, $\int_0^2 x dx = \left[\frac{x^2}{2}\right]_0^2 = 2$. Now we need to compute the improper integral:

$$\int_0^\infty e^{-y} dy = \lim_{r \rightarrow \infty} \int_0^r e^{-y} dy = \lim_{r \rightarrow \infty} [-e^{-y}]_0^r = \lim_{r \rightarrow \infty} -e^{-r} + e^{-0} = 1. \text{ So } I = 2.$$

- ④ Consider the system of functions $F = (f_1, f_2)$ where $f_1(x, y) = x^2 + y^2$ and $f_2(x, y) = e^x \sin(xy)$. Compute its Jacobian.

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} = \begin{bmatrix} 2x & 2y \\ e^x \sin(xy) + e^x \cos(xy) y & e^x \cos(xy) x \end{bmatrix}$$

- ⑤ Show that the polynomial $P(x) = x^3 + 2x^2 - x - 1$ has 3 roots in $[-3, 1]$.

$P(-3) = -27 + 18 + 3 - 1 < 0$
 $P(-1) = -1 + 2 - 1 - 1 = -1 > 0$
 $P(0) = -1 < 0$
 $P(1) = 1 + 2 - 1 - 1 = 1 > 0$

Since P is a polynomial, it's a continuous function. Given that $P(-3) \cdot P(-1) < 0$, by Bolzano's Theorem (or Intermediate Value Theorem), we know $\exists c_1 \in (-3, -1)$ s.t. $P(c_1) = 0$.
 Let's apply the same reasoning with $P(-1) \cdot P(0) < 0$, so another root in $(-1, 0)$.
 $P(0) \cdot P(1) < 0$ so another root in $(0, 1)$. That makes 3 roots.

- ⑥ Let $T(x, y, z) = (x + y, y + z)$ be a map from $\mathbb{R}^3$ to $\mathbb{R}^2$.

a) Is the map linear? Why?

Yes, it is, since T is a matrix map with standard matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$, and matrix maps are linear.

b) Give a basis for $\text{Ker } T$. What is the dimension of $\text{Ker } T$?

Since T is a matrix map, $\text{Ker } T = \text{Nul } A$, so we need to find the solutions to $Ax = 0$, so $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
 so $\begin{cases} x - z = 0 \\ y + z = 0 \end{cases} \Rightarrow \begin{cases} x = z \\ y = -z \end{cases} \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = z \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$, so $\text{Ker } T = \text{Span}\left\{\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}\right\}$ and $\dim \text{Ker } T = 1$.

c) Without finding a basis, which is the dimension of $\text{Im} T$?

Since T is a matrix map and by the Rank Theorem, $\dim \text{Nul} A + \dim \text{Col} A = n = \text{cols. in } A \rightarrow \dim \text{Col} A = \dim \text{Im} T = 3 - 1 = 2$.

d) Find a basis for $\text{Im} T$.

Again, since T is a matrix map, $\text{Im} T = \text{Col} A$, which is spanned by the pivot columns of A , which from part b) we see are the first two, so

$$\text{Im} T = \text{Span}\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$$

⑦ $B = \left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$, $B' = \left\{\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \end{bmatrix}\right\}$ bases for $\mathbb{R}^3$.

a) Find $P_{B \leftarrow B'}$.

We know that $[b_1 \ b_2 \ | \ b'_1 \ b'_2] \sim [I \ | \ P_{B \leftarrow B'}]$ so

$$\left[\begin{array}{cc|cc} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 1 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array} \right] \xrightarrow{P_{B \leftarrow B'}} \left[\begin{array}{cc|cc} 1 & 2 & -1 & 1 \\ 1 & -2 & -1 & 1 \\ 0 & 1 & 2 & 0 \end{array} \right]$$

b) Given a vector v with coordinates in the basis B' $(4, 5, 6)$, find its coordinates relative to B .

Since $[v]_B = P_{B \leftarrow B'} [v]_{B'}$ and we know $[v]_{B'} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$, $[v]_B = \begin{bmatrix} 1 & 2 & -1 \\ 1 & -2 & -1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 4+10-6 \\ 4-10-6 \\ 5+12 \end{bmatrix} = \begin{bmatrix} 8 \\ -12 \\ 17 \end{bmatrix}$.

c) Using $P_{B \leftarrow B'}$, find $P_{B' \leftarrow B}$.

To get $P_{B' \leftarrow B}$ we can invert $P_{B \leftarrow B'}$, so $[P_{B \leftarrow B'} \ | \ I] \sim [I \ | \ P_{B' \leftarrow B}]$, since $P_{B \leftarrow B'} = (P_{B' \leftarrow B})^{-1}$.

$$\left[\begin{array}{cc|cc} 1 & 2 & -1 & 1 \\ 1 & -2 & -1 & 0 \\ 0 & 1 & 2 & 0 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 2 & -1 & 1 \\ 0 & 4 & 0 & 1 \\ 0 & 1 & 2 & 0 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 2 & -1 & 1 \\ 0 & 1 & 0 & 1/4 \\ 0 & 1 & 2 & 0 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 2 & -1 & 1 \\ 0 & 1 & 0 & 1/4 \\ 0 & 0 & -2 & 1/2 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 2 & -1 & 1 \\ 0 & 1 & 0 & 1/4 \\ 0 & 0 & 1 & -1/8 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 2 & 0 & 7/8 \\ 0 & 1 & 0 & 1/4 \\ 0 & 0 & 1 & -1/8 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 0 & 0 & 3/8 \\ 0 & 1 & 0 & 1/4 \\ 0 & 0 & 1 & -1/8 \end{array} \right]$$

$$\sim \begin{bmatrix} 3 & 5 & 4 \\ 2 & -2 & 0 \\ -1 & 1 & 4 \end{bmatrix} \text{ so } P_{B' \leftarrow B} = \frac{1}{8} \begin{bmatrix} 3 & 5 & 4 \\ 2 & -2 & 0 \\ -1 & 1 & 4 \end{bmatrix}$$

⑧ Let V be the vector space of all 2×2 matrices. Show that the subset H of diagonal matrices is a subspace.

The elements of H are of the form $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ with $a, b \in \mathbb{R}$. So we need to prove three things:

- The 0 is in H . It is true since the 0 matrix is also diagonal, since $0 \in \mathbb{R}$.
- If $A, B \in H$ then $A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ and $B = \begin{bmatrix} c & 0 \\ 0 & d \end{bmatrix}$, $A+B = \begin{bmatrix} a+c & 0 \\ 0 & b+d \end{bmatrix}$ which is a diagonal matrix, so $A+B \in H$.
- If $A \in H$ and $c \in \mathbb{R}$, then $cA = \begin{bmatrix} ca & 0 \\ 0 & cb \end{bmatrix} = \begin{bmatrix} ca & 0 \\ 0 & cb \end{bmatrix}$, which is a diagonal matrix, so $cA \in H$.

⑨ $A = \begin{bmatrix} 7 & 0 & -4 \\ 0 & 5 & 0 \\ 5 & 0 & -2 \end{bmatrix}$.

a) Find all eigenvalues of A .

For that we solve $\det(A - \lambda I) = 0 \rightarrow \det(A - \lambda I) = (5 - \lambda) \begin{vmatrix} 7 - \lambda & -4 \\ 5 & -2 - \lambda \end{vmatrix} = (5 - \lambda)((7 - \lambda)(-2 - \lambda) + 20) = (5 - \lambda)(-14 - 7\lambda + 2\lambda^2 + 20) = (5 - \lambda)(\lambda^2 - 5\lambda + 6) = 0$

$$\lambda_1 = 5 \quad \lambda = \frac{1}{2} \frac{5 \pm \sqrt{25 - 4 \cdot 6}}{2} = \frac{5 \pm 1}{2} = \begin{cases} \lambda_2 = 3 \\ \lambda_3 = 2 \end{cases}$$

b) Give a basis for every eigenspace.

Let's start with $\lambda_1 = 5$. We need to solve $(A - 5I)x = 0$.

$$\begin{bmatrix} 2 & 0 & -4 & 0 \\ 0 & 0 & 0 & 0 \\ 5 & 0 & -7 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 0 & -7 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 0 & -7 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{cases} x_1 - 2x_3 = 0 \\ x_3 = 0 \end{cases} \rightarrow \begin{cases} x_1 = 0 \\ x_3 = 0 \end{cases} \rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \text{ so basis } \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

For $\lambda_2 = 3$:

$$\begin{bmatrix} 4 & 0 & -4 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{cases} x_1 - x_3 = 0 \\ x_2 = 0 \end{cases} \rightarrow \begin{cases} x_1 = x_3 \\ x_2 = 0 \end{cases} \rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \text{ basis } \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

For $\lambda_3 = 2$:

$$\begin{bmatrix} 5 & 0 & -4 & 0 \\ 0 & 3 & 0 & 0 \\ 5 & 0 & -4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -4/5 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{cases} x_1 - 4/5 x_3 = 0 \\ x_2 = 0 \end{cases} \rightarrow \begin{cases} x_1 = 4/5 x_3 \\ x_2 = 0 \end{cases} \rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 4/5 \\ 0 \\ 1 \end{bmatrix}, \text{ so basis } \left\{ \begin{bmatrix} 4/5 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

c) Is A diagonalizable?

Yes, since all eigenvalues are different. $D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, $P = \begin{bmatrix} 0 & 1 & 4/5 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$.

10) Let $W = \text{Span} \left\{ \underbrace{\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}}_{a_1}, \underbrace{\begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix}}_{a_2}, \underbrace{\begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix}}_{a_3} \right\}$ be a subspace of $\mathbb{R}^3$.

a) Find an orthogonal basis for W .

Since the spanning set is linearly independent and not orthogonal, we can apply the Gram-Schmidt process.

$b_1 = a_1$

$$b_2 = a_2 - \frac{a_2 \cdot b_1}{b_1 \cdot b_1} b_1 = \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix} - \frac{8}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$$

$$b_3 = a_3 - \frac{a_3 \cdot b_1}{b_1 \cdot b_1} b_1 - \frac{a_3 \cdot b_2}{b_2 \cdot b_2} b_2 = \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix} - \frac{-4}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \frac{12}{6} \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, \text{ so the basis is } \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \right\}.$$

11) True or false?

a) Since $\{1, t, t^2\}$ has 3 elements and is linearly independent, it is a basis for P_3 .

False. The set is a basis for P_2 . For P_3 we need $\{1, t, t^2, t^3\}$.

b) $2 \det(A+B) = 2 \det A + 2 \det B$

False. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, $A+B = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$, $\det A = 1$, $\det B = 4$, $\det(A+B) = 9$, $2 \det(A+B) = 18 \neq 2 \det A + 2 \det B = 2 + 8 = 10$.

c) If u, v are orthogonal, $\|u+v\|^2 = \|u\|^2 + \|v\|^2$.

True. $\|u+v\|^2 = (u+v) \cdot (u+v) = u \cdot u + u \cdot v + v \cdot u + v \cdot v = \|u\|^2 + \|v\|^2 + 2u \cdot v$. Since u, v are orthogonal, $u \cdot v = 0$, so the result follows.